

Verification summary

MQ9 prototype fabrication package / Revision 1 / 19 September 2026

Check	Result and scope
Source inventory	62 original assembly occurrences retained for traceability; three tiny slivers excluded from fabrication. Assembly placements recovered independently of separated-part positions.
Plywood	22 profiles /31 pieces. Missing opposite-wing parts added. Native millimetre DXFs audit clean. Revised plate solids and source differences are recorded; revised blanks are not falsely labelled identical to tapered originals.
Plywood sheets / PDF	Three stock layouts; no bounding-box overlap.100 x 20 mm calibration verified from PDF vectors. Optimized plywood paper set uses 7 pages. Paired registration discrepancy at most 0.0000371 mm; sampled curve-scale discrepancy at most 0.0000717 mm.
Foam development	33 patterns /35 blanks. Maximum sampled mesh-edge strain 0.3664%; no reversed triangles or self-crossing boundaries. Absolute 0.2 mm mesh, checked against exact STEP surfaces. Maximum sampled normal departure 0.19971 mm. This is numerical geometry, not a material test.
Foam seams / sheets	Largest checked introduced-seam length mismatch 0.02667 mm. Three 1000 x 600 mm sheet layouts fit 35 blanks with no bounding-box overlap.Optimized foam paper set uses 23 pages including index/calibration. True-size coordinates and paired registration were rechecked; single-page-fit patterns remain whole.
Aircraft print meshes	11 meshes plus one separate test coupon. All aircraft meshes closed and consistently wound; manifold library reports NoError. Absolute export tolerance 0.015 mm. Maximum CAD/mesh volume discrepancy 0.05194%. Adaptive integration with BSpline spans avoids earlier misleading volumes.
Changed mating geometry	Zero computed volumetric overlap at extended nosewire/bearing, wire/mount, wire/spine, frontjoiner/sleeve, joiner/root, sleeve/middle rib and revised tailfairing/bulkhead. Typical nominal tube radial clearance 0.10 mm. Other local foam contact is explicitly handled by forming/relief instructions.

What changed from the source

1. Added the missing mirrored wing ribs, spars, end plates and rear joiner support; quantities refer to a full aircraft.
2. Replaced the contradictory front tube references with a centred 8/6 mm joiner and paired 10/8.2 mm sleeves. Recentred the rear 8/6 mm tube; omitted the redundant short source stub.
3. Moved the middle-rib sleeve bore onto the joiner axis and set 10.2 mm sleeve holes. Set 8.2 mm root/rear bores; physical tube tolerances remain a matched-fit shop check.
4. Combined overlapping centre spines. Split that spine deliberately at the nose shaft and added two 2.5 mm cheek plates, four M2 through-bolts and matching printed mount pockets. Extended the nosewire through its bearing/steering arm.
5. Changed motor holes from 25 mm to 19 mm pitch circle for the verified Emax drawing. Added/enlarged main-gear through-bolt holes for M3 fasteners.
6. Prepared prismatic wood blanks where the source had small side tapers. Sanding references and source departures are declared. Relieved the aft-hoop receiving notch without severing the spine.
7. Cleared the printed tailfaring bulkhead pocket and nose steering bore. Re-exported all selected print meshes with absolute, rather than relative, deflection.
8. Developed curved foam surfaces using a preserved outside-paper process for 5 mm FliteBoard. Tail controls are two-layer 10 mm blanks. Recomputed patterns on a finer absolute mesh and provided seam/relief instructions.

Operations still required at the workshop

Measure stock and run slot/print/foam coupons. Dry-fit all neighbours together. Finish prescribed plywood edges and foam inner-core relief; cut gear exits from actual installed geometry. Test printed moving fits and fastener stacks. Form metal gear using suitable material and finite tooling bend radii. Check structural joints and gear deflection for the actual completed mass. Record finished mass and balance.

The owner must then verify exact unidentified electronics connectors, servo voltage/current, control direction, RC failsafe, motor/propeller load and ground clearance. The provisional 522 mm CG mark is not a flight-approved range. No physical load, material forming, aerodynamic or flight test has been claimed.

An independent reviewer checked the electronics guide and flagged the short source nosewire, stale earlier notes and physical fit limits. The revised wire and current guide address those findings; earlier packages and Rev 0 notes are archived outside this ZIP.